

Banking Customer Churn & Credit Card Analytics

End-to-End Data Analyst Portfolio Project — SQL • Excel • Power BI

Project scope. Synthetic banking data was analyzed to understand customer churn, transaction behavior, account balances, branch performance, and credit-card utilization. The project demonstrates data validation, SQL analysis, Excel validation, relational modeling, DAX, dashboard design, and business communication.

Dataset	Size
Customers	1,000
Accounts	1,450
Transactions	12,015
Credit cards	720
Branches	20

Key findings

- Overall customer churn rate: **19.10%**.
- Total account balance: approximately **162.9M**; Savings holds the largest aggregate balance.
- Successful transactions: **11,318** out of 12,015 total transactions.
- Overall transaction failure rate: **3.92%**; Branch and POS are highest at about 4.52%.
- Successful transaction amount: approximately **50.1M**.
- Portfolio credit utilization: **47.11%**.
- High-utilization-card rate (>70%): **26.11%**.
- Active cards: **91.94%** of all cards.

Analytical approach

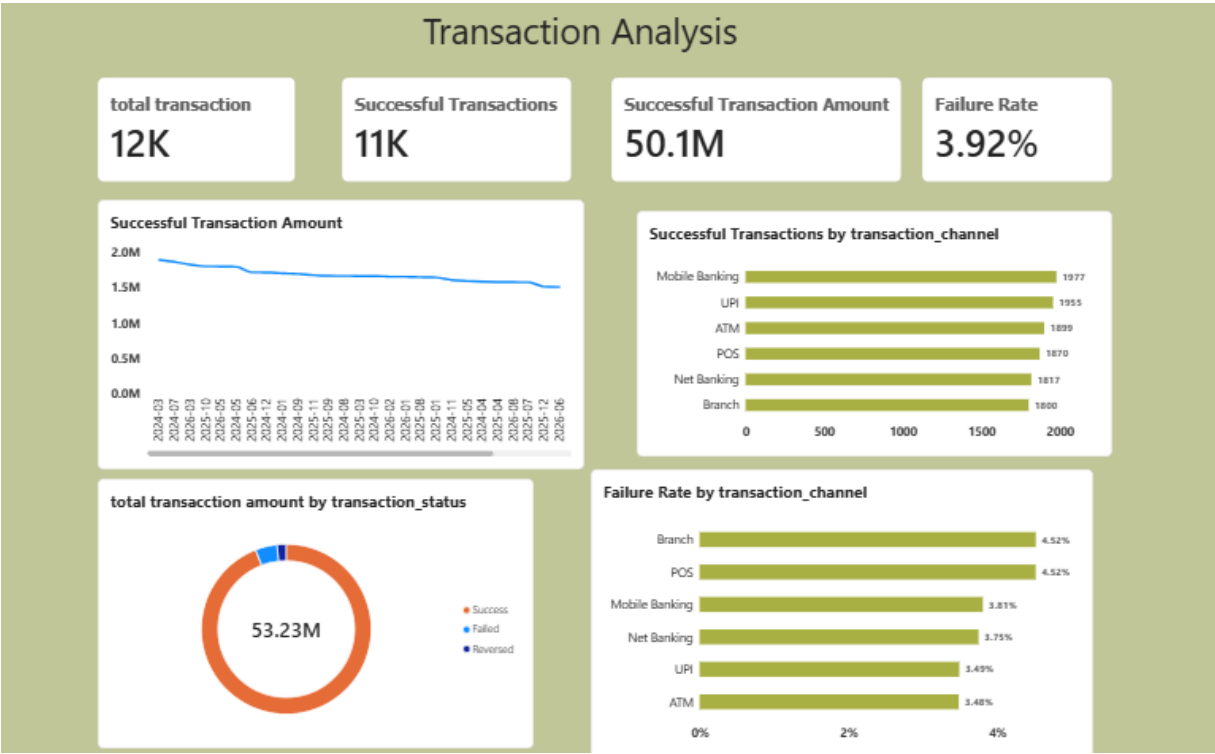
SQL: data-quality checks, joins, CTEs, window functions, inactivity, segmentation, channel analysis, branch analysis, Pareto contribution, and monthly growth. **Excel:** data-quality formulas and PivotTables used to validate churn and transaction metrics. **Power BI:** star-like relational model, DAX measures, calculated columns, slicers, KPI cards, bar/donut/line visuals, and three report pages.

Dashboard pages

Customer Churn Analysis



Transaction Analysis



Account & Credit Analysis

Account & Credit Analysis

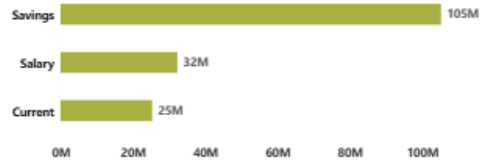
Total Account Balance
162.9M

total cards
720

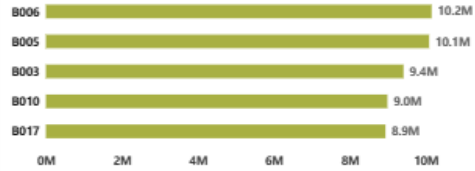
Credit Utilization %
47.11%

High Utilization Rate
26.11%

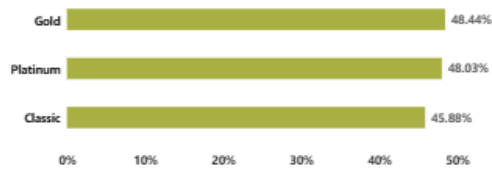
Balance by account_type



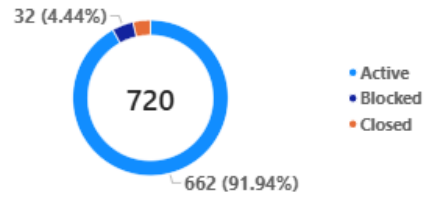
Branch Performance



Credit Utilization by Card Type



total cards by card_status



Business recommendations

- Investigate high-churn segments while validating segment size and avoiding causal claims from descriptive analysis.
- Review Branch and POS transaction journeys because their observed failure rates are higher than other channels.
- Monitor customers/cards above the 70% utilization threshold for proactive engagement and risk management.
- Use branch balance concentration as an input to service and relationship-management planning.
- Track these KPIs over time and add retention/cohort analysis when production history is available.

Portfolio talking point

"I built an end-to-end banking analytics project using SQL, Excel, and Power BI. I focused heavily on data granularity so one-to-many relationships did not inflate customer metrics. I validated the analysis in Excel, created DAX measures in Power BI, and built three business-focused dashboard pages covering churn, transactions, and account/credit behavior."

Dataset note: synthetic data created for portfolio and interview practice.